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Subject Code: RCA1041

Roll No:

MCA-INT
(SEM IX) THEORY EXAMINATION 2024-25
DISTRIBUTED DATABASE SYSTEMS

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief.

$$2 \times 07 = 14$$

Q no.	Question
a.	What is a transaction in a database management system (DBMS)?
b.	Define a serial schedule.
c.	What is the role of a lock manager in a lock-based protocol?
d.	Discuss data fragmentation in a distributed database system?
e.	How does atomicity affect recovery in distributed systems?
f.	What is a semi-join in distributed query processing?
g.	How does shadow paging contribute to database recovery?

SECTION B

2. Attempt any *three* of the following:

07 x 3 = 21

Q no.	Question
a.	Compare and contrast conflict serializability with view serializability, highlighting the scenarios in which each would be applicable.
b.	Explain the working of the Thomas Write Rule in timestamp-based protocols. What role does it play in ensuring serializability and conflict resolution?
c.	Discuss the concept of distributed locking schemes in a distributed database system. How do they ensure the consistency and isolation of transactions?
d.	Describe the process of log-based recovery in database systems. How do write-ahead logs help in ensuring the durability of transactions?
e.	Discuss the main protocols for distributed deadlock detection. How do they differ in terms of resource allocation, message passing, and handling of distributed cycles in a system?

SECTION C

3. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question
a.	Explain the concept of a recoverable schedule. Why is it important for ensuring transaction consistency in case of failures?
b.	Explain the concept of deadlock in concurrent transaction processing and the strategies used to resolve it.

4. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

Q no.	Question
a.	Discuss the multi-version concurrency control (MVCC) technique. How does it handle concurrent transactions, and how does it differ from traditional locking mechanisms?
b.	Describe the architecture of a locking scheduler in a DBMS. How does it coordinate transactions and manage the allocation of locks?



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DISTRIBUTED DATABASE SYSTEMS

TIME: 3 HRS**M.MARKS: 70****07 x 1 = 07****5. Attempt any one part of the following:**

Q no.	Question
a.	Discuss the role of the MOSS protocol in distributed database systems. How does it handle transaction conflicts and ensure serializability in a highly concurrent environment?
b.	Discuss the impact of long-duration transactions on resource management, locking, and transaction scheduling in distributed systems. How do these transactions affect the overall throughput of the system?

07 x 1 = 07**6. Attempt any one part of the following:**

Q no.	Question
a.	Explain how recovery works in message-passing systems. How do these systems ensure reliable communication and recovery from failures such as message loss or duplication?
b.	Discuss the challenges posed by orphan and inconsistent messages during the recovery phase in message-passing systems. How do these messages affect the overall system state? https://www.aktuonline.com

07 x 1 = 07**7. Attempt any one part of the following:**

Q no.	Question
a.	Explain the challenges of performing multiway joins in a distributed environment, including network latency, data partitioning, and load balancing. How can these challenges be overcome?
b.	Explain different strategies for updating replicated data in distributed systems. How do these strategies ensure consistency and fault tolerance in the presence of failures?